$\underline{r} = \underline{r}(t)$ is pos. vect. rel. to O

$\underline{v} = \underline{v}(t) = \frac{\partial \underline{r}(t)}{\partial t}$ is vel. vect.

$v = v(t) = |\underline{v}(t)| = \sqrt{\underline{v}(t)\cdot\underline{v}(t)}$ is speed

$\underline{a} = \underline{a}(t) = \frac{\partial \underline{v}(t)}{\partial t} = \frac{\partial^2 \underline{r}(t)}{\partial t^2}$ is accel. vect.

$s = s(t)$ is tot. dist. since $t=0$

$\frac{\partial s(t)}{\partial t} = |\underline{v}|$, $s(t_2) - s(t_1) = \int_{t_1}^{t_2} \frac{\partial s}{\partial \tau} d\tau = \int_{t_1}^{t_2} |\underline{v}(\tau)| d\tau$

$\underline{r}(t) = \underline{r}(s(t))$, $\underline{r}$ diff wrt t diff wrt s, def. v and a

$v = \dot{\underline{r}} = \frac{\partial \underline{r}}{\partial s}\cdot\frac{\partial s}{\partial t} = \frac{\partial \underline{r}}{\partial s}\cdot v$

$a = \dot{v}\frac{\partial \underline{r}}{\partial s} + v\frac{\partial}{\partial t}\left(\frac{\partial \underline{r}}{\partial s}\right) = \dot{v}\frac{\partial \underline{r}}{\partial s} + v^2$ ($\frac{\partial}{\partial t}\left(\frac{\partial \underline{r}}{\partial s}\right) = \frac{\partial \underline{r}}{\partial s}\cdot v$)

even if speed is constant, can still have non-zero acceleration if $v \neq 0$ and the path is not straight $\frac{\partial^2 \underline{r}}{\partial s^2} \neq 0$ (circ. motion)

does $s(t) = x(t)$? → no, not gen. $s(t)$ meas. d along path, $x(t)$ is hor. coord. = only for motion along x-axis in $\mathbb{R}$ w/ same t_0

circ. motion @ const. speed

$\underline{r}(t) = R_0(\cos(\omega t)\underline{e}_x + \sin(\omega t)\underline{e}_y)$

$v(t) = |\underline{v}(t)| = |\omega| R_0$

$\underline{a}(t) = \ddot{\underline{r}}(t) = -\omega^2 \underline{r}$

$\underline{r} = x\underline{e}_x + y\underline{e}_y + z\underline{e}_z = r\underline{e}_r + z\underline{e}_z$, using $\{\underline{e}_r, \underline{e}_\theta, \underline{e}_z\}$ instead of $\{\underline{e}_x, \underline{e}_y, \underline{e}_z\}$ as basis. it's cylindrical basis. these aren't fixed but depend of position (θ).

$\dot{\underline{e}}_r = \left(\partial \underline{e}_r/\partial\theta\right)\dot{\theta} = \dot{\theta}\underline{e}_\theta$, $\dot{\underline{e}}_\theta = \left(\partial \underline{e}_\theta/\partial\theta\right)\dot{\theta} = -\dot{\theta}\underline{e}_r$ ($\dot{r} = \ddot{r} = 0$)

Consider circular motion w/ some $\theta(t)$, $r = R_0 = $ const

$\underline{r} = R_0\underline{e}_r$, $\underline{v} = R_0\dot{\theta}\underline{e}_\theta$, → $v = |\underline{v}|^2 = R_0^2\dot{\theta}^2$

$\underline{a} = -R_0\dot{\theta}^2\underline{e}_r + R_0\ddot{\theta}\underline{e}_\theta = -\underbrace{\frac{R_0^2\dot{\theta}^2}{R_0}}_{=v^2/R_0}\underline{e}_r + \frac{\partial}{\partial t}\left(R_0\dot{\theta}\right)\underline{e}_\theta$ ($= \begin{cases} v \text{ if } \dot{\theta} > 0 \\ -v \text{ if } \dot{\theta} < 0 \end{cases}$)

$= -\frac{v^2}{R_0} + \dot{v}\,\mathrm{sign}(\dot{\theta})\underline{e}_\theta$

at a slalom track: $a_t = \dot{v}$, $a_n = -Kv^2$. $K(x) = |y''(x)|/\left(1 + (y')^2\right)^{3/2}$

$= \frac{|b^2 A \sin bx|}{(1 + b^2A^2\cos^2 bx)^{3/2}}$, at peak $x = \pi/2b \to K = b^2A$, $a_n = -b^2Av^2$

$\underline{a} = \dot{v}\underline{e}_x - b^2Av^2\underline{e}_y$

If F_{wall} is exponential, why doesn't my elbow tear when I fling the ball? Unless tube is huge, ball exits ρ quickly so that torque on joint ($-F_{wall}\, r$) never gets too big

if $m\ddot{y} = -\frac{2T_y}{\sqrt{(D/2)^2 + y^2}} - mg$, $\ddot{y} = v\frac{\partial v}{\partial y}$

then $\int_0^{v_t} mv\,dv = \int_{-L}^{0} -\left(\frac{2T_y}{\sqrt{(D/2)^2 + y^2}} - mg\right)dy$

Friction: to complete formula for F_f when $|\underline{v}_S| = 0$, 1. if $|\underline{v}_S| = 0$ and $|\underline{F}_{trial}| \leq \mu_S F_n$ then $\underline{F}_f = \underline{F}_{trial}$. 2. if $|\underline{v}_S| = 0$ and $|\underline{F}_{trial}| > \mu_S F_n$ then $\underline{F}_f = \mu_S F_n\left(\underline{F}_{trial}/|\underline{F}_{trial}|\right)$

$W_{AB} = \int_{t_A}^{t_B} P\,dt = \int_{t_A}^{t_B} \underline{F}\cdot\underline{v}\,dt = \int_{t_A}^{t_B} \underline{F}\cdot\frac{\partial \underline{r}}{\partial t}dt = \int_{path} \underline{F}\,\partial\underline{r}$

$K = \frac{1}{2}m\underline{v}\cdot\underline{v} = \frac{1}{2}mv^2 \to \dot{K} = \frac{\partial}{\partial t}\left(\frac{1}{2}m\underline{v}\cdot\underline{v}\right) = \sum_i \left(\underline{F}^i\cdot\underline{v}\right) \to W_{AB}^{tot} = \sum_i \int_{t_A}^{t_B} \underline{F}^i\cdot\underline{v}\,dt = K_B - K_A$

dissipative force: $\underline{F}\cdot\underline{v} < 0$ when $|\underline{v}| \neq 0$

powerless forces: $\underline{F}\cdot\underline{v} = 0$, speed never changes, $P = F\cdot v$

work done by a conservative force depends only on the PE of the force at the path's endpoints. Path has no effect



B/c B spins "with" the ball, it should look stationary to B, $\underline{v}_B = 0$, and located at $\underline{r}^B = L\underline{e}_x$ always, where $\underline{e}_x$ always means "rightward" and $\underline{e}_y$ always means "upward". However, A sees $\underline{r}^A = L(\cos\omega t\,\underline{e}_x + \sin\omega t\,\underline{e}_y)$

$\underline{r}^A = \underline{R}(\phi)\underline{r}^B$, $\underline{v}^A = \underline{R}(\phi)\left(\underline{v}^B + \dot{\phi}\underline{e}_z \times \underline{r}^B\right)$

$\underline{a}^A = \underline{R}(\phi)\left(\underline{a}^B + \ddot{\phi}\underline{e}_z \times r^B - \dot{\phi}^2(\underline{r}^B - z^B\underline{e}_z)\right) + 2\dot{\phi}\underline{e}_z \times \underline{v}^B$

inertial frames are deemed to be moving at a const. vel. rel. to universe. Frame of obs. A is inertial, what about frame B:

$m\underline{a}^B = \sum \underline{F}^B - m\ddot{\phi}\underline{e}_z \times \underline{r}^B + m\dot{\phi}^2(\underline{r}^B - z^B\underline{e}_z) - 2m\dot{\phi}\underline{e}_z \times \underline{v}^B$

(Euler force; centrifugal force; coriolis force)

(Ang. vel)

$\underline{a} = -\beta\underline{r}/|\underline{r}|^3$, in terms of energy bal, $\underline{F}_g$ is conservative with $U_g = -\beta m/|\underline{r}|$. there are no ne forces so energy is conserved:

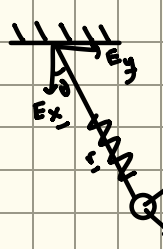
$\frac{1}{2}mv^2 - \beta m/|\underline{r}| = E = $ constant

ang. momentum is fixed

Kepler's 2nd law - $(\text{Area}_{t_1\to t_2})/(t_2 - t_1)$ is fixed const. for each orbit regardless of how t_1, t_2 is picked. "= areas take = time"

Speed of m in circular orbit $= \frac{2\pi R}{T_{orbit}} = \sqrt{\frac{GM}{R}}$. So if we shot a bullet on earth, it would have speed $\sqrt{\frac{GM}{R_E}}$. $v_{esc} = \sqrt{\frac{2GM}{R_E}}$. Thus KE to orbit earth is half that needed to esc. earth.

friction is case where $F_{AB} = -F_{BA}$ does not point || to $\underline{r}_B - \underline{r}_A$ (not grav, elec force, tens, spri)



$\underline{E}_z = \underline{e}_r \times \underline{e}_\theta = \underline{E}_x \times \underline{E}_y$

$\underline{r} - \underline{r}_A = r\underline{e}_r - \underline{0} = r\underline{e}_r$

$\frac{\underline{r}-\underline{r}_A}{\|\underline{r}-\underline{r}_A\|} = \frac{r\underline{e}_r}{|r|} = \underline{e}_r$

$\underline{F}_s = k(r-l_0)(-\underline{e}_r)$

both springs are unstretched when $y=0$, $l = L$

$\underline{r} - \underline{r}_A = L\underline{E}_x + y\underline{E}_y$

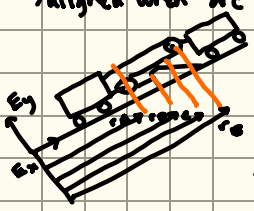
$\|\underline{r}-\underline{r}_A\| = \sqrt{L^2+y^2}$ $\quad l = \sqrt{L^2+y^2}$

$F_{s_1} = k(l-L)\frac{L\underline{e}_x - y\underline{e}_y}{l}$

$-2k(1-\frac{L}{l})y\underline{e}_y = m\ddot{y}\underline{e}_y$

n-t basis: $\underline{a} = \dot{v}\underline{e}_t + \frac{v^2}{\rho}\underline{e}_n$

↳ aligned when arc length is oriented WRT to origin



$l_1 = l_2$, $L = l_1 + l_2 + l_3$

$\dot{L} = 0 = 2\dot{l}_1 + \dot{l}_3$

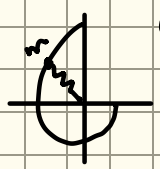
$v_A = v_A\underline{e}_x = -\dot{l}_3\underline{e}_x$

$v_B = v_B\underline{e}_x = (-\dot{l}_3 + \dot{l}_1)\underline{e}_x$

$\underline{e}_t = \frac{1}{\sqrt{1+m(\theta)^2}}\underline{e}_x + \frac{m(\theta)}{\sqrt{1+m(\theta)^2}}\underline{e}_y$

$\underline{e}_n = \frac{m(\theta)}{\sqrt{1+m(\theta)^2}}\underline{e}_x - \frac{1}{\sqrt{1+m(\theta)^2}}\underline{e}_y$

$\underline{e}_r = \cos\theta\underline{e}_x + \sin\theta\underline{e}_y$, $\underline{e}_\theta = -\sin\theta\underline{e}_x + \cos\theta\underline{e}_y$



$N = N\underline{e}_n$, $W = -mg\underline{e}_y$, $F_s = -k(\|\underline{r}-\underline{r}_0\| - l_0)\frac{\underline{r}-\underline{r}_0}{\|\underline{r}-\underline{r}_0\|}$

$\underline{a} = \dot{v}\underline{e}_t - k(\theta)v^2\underline{e}_n$, $k(\theta) = 1/\rho$



$\dot{\underline{e}}_r = \dot{\theta}\underline{e}_\theta$, $\dot{\underline{e}}_\theta = -\dot{\theta}\underline{e}_r$

$F = -d(r-(r-e_y)\underline{e}_y)$

$\underline{r} = x\underline{e}_x + y\underline{e}_y$

$\underline{v} = v\underline{e}_t$

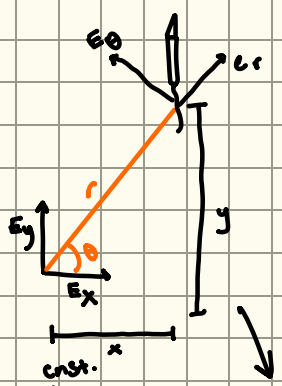
$\underline{a} = \dot{v}\underline{e}_t - \frac{v^2}{\rho}\underline{e}_n$ $\quad W = mg\underline{e}_z$

$N = N_z\underline{e}_z + N_n\underline{e}_n$

$F_s = -k(x\underline{e}_x + y\underline{e}_y)$

$F = -d\dot{x}\underline{e}_x$

$E = \frac{1}{2}mv^2 + mgz + \frac{1}{2}k(x^2+y^2) + \frac{1}{2}dx^2$



$r = x\underline{E}_x + y\underline{E}_y = r\underline{e}_r = 9000\underline{e}_r = 4500\underline{e}_x + 7794.2\underline{e}_y$

$v = \dot{y}\underline{E}_y = \dot{r}\underline{e}_r + r\dot{\theta}\underline{e}_\theta$, $v_\theta = r\dot{\theta} = 180$, $v_\theta = v\cos 60 \Rightarrow v = 360$

$a = \ddot{y}\underline{E}_y = \ddot{r}\underline{e}_r + \dot{r}\dot{\theta}\underline{e}_\theta + \dot{r}\dot{\theta}\underline{e}_\theta + r\ddot{\theta}\underline{e}_\theta - r\dot{\theta}^2\underline{e}_r$

$\dot{r} = v\sin 60 = 311.8$, $\dot{\underline{e}}_r = \dot{\theta}\underline{e}_\theta$, $\dot{\underline{e}}_\theta = -\dot{\theta}\underline{e}_r$

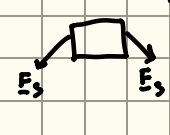
$a_r = \ddot{r} - r\dot{\theta}^2 = 21 - 9000(0.02)^2 = 17.4$, $a_r = a\sin 60$

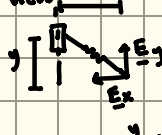
$a = 20.1$

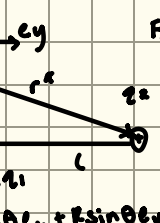
$\underline{r} = R\cos\theta\underline{e}_x + R\sin\theta\underline{e}_y$

$\underline{v} = -R\dot{\theta}\sin\theta\underline{e}_x + R\dot{\theta}\cos\theta\underline{e}_y$

$\underline{a} = -R\ddot{\theta}\sin\theta\underline{e}_x - R\dot{\theta}^2\cos\theta\underline{e}_x + R\ddot{\theta}\cos\theta\underline{e}_y - R\dot{\theta}^2\sin\theta\underline{e}_y$







$\underline{F}_T = -T\cos\theta_0\underline{e}_x - T\sin\theta_0\underline{e}_y$, powerless, non-cons. $\quad \underline{F}_T \cdot \underline{v} = 0$

$\underline{F}_g = m_1g\underline{e}_x$, cons. $\quad \underline{F}_d = -c\sin\theta_0\underline{e}_x + c\cos\theta_0\underline{e}_y$

$\underline{F}_l = -\frac{k}{l}\underline{e}_y$, cons. $\quad$ drag $= -cR(\theta_0 - \theta)\uparrow$ non-conserv.

$K_i + U_i + W_{nc} = K_f + U_f$ ↑ spring ↑ friction

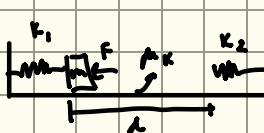
$\underline{r}_A = \cos\alpha\underline{e}_x + \sin\alpha\underline{e}_y$

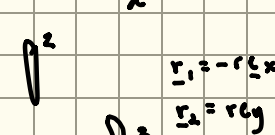
$\underline{r}_B = R\cos\alpha\underline{e}_x + R\sin\alpha\underline{e}_y$

$\underline{s}_A = -k(\|\underline{r}_A - \underline{r}_B\| - l_0)\frac{\underline{r}_A - \underline{r}_B}{\|\underline{r}_A - \underline{r}_B\|}$

$F_s = -k(r-l_0)\frac{r}{r}$, $F_{cf} = m\Omega^2 r$

$F_n = Nn$, $F_{cor} = -2m\Omega\underline{e}_x \times \underline{v}$

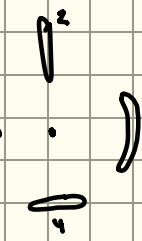




$r = x\underline{e}_x$, $v = \dot{x}\underline{e}_x$, $a = \ddot{x}\underline{e}_x$

$F_1 = -k_1(\|l - \Delta l_1\underline{e}_x - (-l_1\underline{e}_x)\| - l_1)\frac{\Delta l_1\underline{e}_x - (-l_1\underline{e}_x)}{\|\Delta l_1\underline{e}_x - (-l_1\underline{e}_x)\|}$

$F_2 = -k_2(\|(d+\Delta l_2)\underline{e}_x - (d+l_2)\| - l_2)\frac{(d+\Delta l_2)\underline{e}_x - (d+l_2)}{\|(d+\Delta l_2)\underline{e}_x - (d+l_2)\|}$



$\underline{r}_1 = -r\underline{e}_x$	$\underline{v}_1 = -u\underline{e}_x$	$\underline{a}_1 = a_1\underline{e}_x$	$\underline{N}_1 = N_1\underline{e}_y$
$\underline{r}_2 = r\underline{e}_y$	$\underline{v}_2 = -u\underline{e}_y$	$\underline{a}_2 = a_2\underline{e}_y$	$\underline{N}_2 = N_2\underline{e}_x$
$\underline{r}_3 = r\underline{e}_x$	$\underline{v}_3 = u\underline{e}_y$	$\underline{a}_3 = a\underline{e}_y - \frac{u^2}{r}\underline{e}_x$	
$\underline{r}_4 = -r\underline{e}_y$	$\underline{v}_3 = u\underline{e}_x$	$\underline{a}_4 = a\underline{e}_x$	

$\underline{F}_{co}' = -2m\omega\underline{e}_z \times -u\underline{e}_x$

$= -2m\omega u\underline{e}_x$

$= 2m\omega u\underline{e}_y$

$= -2m\omega u\underline{e}_y$

$F_{cent}' = +m\omega^2 r\underline{e}_x$

$-m\omega^2 r\underline{e}_y$

$-m\omega^2 r\underline{e}_x$

$m\omega^2 r\underline{e}_y$



$-mg\underline{E}_z + N_\theta\underline{e}_\theta + N_z\underline{E}_z - kr\underline{e}_r = m\underline{a}$

$\underline{r} = r\underline{e}_r + z\underline{E}_z$, $\underline{v} = \dot{r}\underline{e}_r + r\dot{\theta}\underline{e}_\theta + \dot{z}\underline{E}_z$

$\underline{a} = (\ddot{r} - r\dot{\theta}^2)\underline{e}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\underline{e}_\theta + \ddot{z}\underline{E}_z$

$\underline{r} = p\underline{E}_p + z\underline{E}_z$, if box does not slip, p and z stay const, however $\underline{e}_p$ rotates so:

$\dot{\underline{E}}_p = w\underline{E}_\theta$, $\dot{\underline{E}}_\theta = -w\underline{E}_p$

so $\underline{v} = pw\underline{E}_\theta$, $\underline{a} = -pw^2\underline{E}_p$

FBD:

$N = -mg\underline{E}_z$

$\underline{N} = N(-\sin\beta\underline{E}_p + \cos\beta\underline{E}_z)$

$\underline{F}_f = f(\cos\beta\underline{E}_p + \sin\beta\underline{E}_z)$

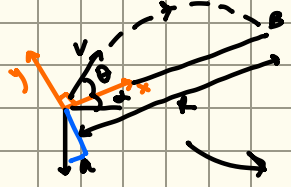
$|f| \le \mu_s N$, $\mu_s = 0.30$

$\sum \underline{F} = m\underline{a} = (-N\sin\beta + f\cos\beta)\underline{E}_p + (N\cos\beta + f\sin\beta - mg)\underline{E}_z$

$= -mpw^2\underline{E}_p$

along $\underline{E}_p$, $\underline{E}_z$, $f_s = \mu_s N$ ↖ uphill

$w_{min} = \sqrt{\frac{g\sin\beta - \mu_s\cos\beta}{p\cos\beta + \mu_s\sin\beta}} = 3.41$ rad/s



$v_{0x} = v_0\cos(\theta - \alpha)$ $\quad v_{y_0} = v_0\sin(\theta - \alpha)$

$a_x = -g\sin\alpha$, $a_y = -g\cos\alpha$

$x(t) = v_0\cos(\theta-\alpha)t - \frac{1}{2}g\sin\alpha t^2$

$y(t) = v_0\sin(\theta-\alpha)t - \frac{1}{2}g\cos\alpha t^2$

max R is when $y = 0$ so:

$0 = v_0\sin(\theta-\alpha)t - \frac{1}{2}g\cos\alpha t^2$

$0 = t(v_0\sin(\theta-\alpha) - \frac{1}{2}g\cos\alpha t)$

$t \neq 0$, $\frac{2v_0\sin(\theta-\alpha)}{g\cos\alpha}$

$R = v_0\cos(\theta-\alpha)\cdot\frac{2v_0\sin(\theta-\alpha)}{g\cos\alpha} - \frac{1}{2}g\sin\alpha\left(\frac{2v_0\sin(\theta-\alpha)}{g\cos\alpha}\right)^2$